

EMPLOYEE PERFORMANCE ANALYTICS

Integrated Enterprise Retention Analysis & Agile Framework Delivery Report

Company: Codec Technologies **Submitted By:** Sthita Pragyan Sahu **Target:** Human Resources Department

1. Executive Summary

This technical assessment details employee productivity, workplace satisfaction, and attrition predictors across a cohort of 40 active profiles at **Codec Technologies**. Using an integrated operational pipeline driven by Excel modeling, SQL structure, and visualization layouts, this project identifies systemic indicators affecting employee satisfaction and corporate retention.

Crucially, this analysis was structured and executed via the **Agile Scrum Methodology** over 1-week sprint iterations. This dual framework ensured continuous tracking of user stories, rapid workflow adjustments through a visual Kanban platform, and full alignment with target strategic deadlines.

2. Agile Development Framework & Execution Metrics

The lifecycle of this analysis strictly adhered to modern product management frameworks to ensure clean extraction-transform-load (ETL) pipelines and dashboard prototyping.

Scrum Project Alignment

- **Sprint Duration:** 1-Week Intervals
- **Product Owner Accountabilities:** HR Manager (KPI rule baseline mapping)
- **Scrum Master & Technical Team:** Sthita Pragyan Sahu (Lead Data Analyst)

Sprint Velocity & Technical Backlog Completion

Sprint Sequence	Sprint Goal	Core Deliverable	Status
Sprint 1	Data Cleaning & Ingestion	Structural cleanup, validation, type casting	100%
Sprint 2	KPI Analysis	Baseline aggregation (performance, satisfaction, hours)	100%
Sprint 3	Visualization Design	Pivot chart mapping & layout drafting	100%
Sprint 4	Insights & Strategy	Retention analysis deep dive & documentation compilation	90%

Current Kanban Board State: All core technical items including *Data Cleaning*, *KPI Computation*, and *Pivot Overviews* reside in **Done**. *Dashboard UI Review* and *Report Sign-off* are situated within the **Testing** phase. *Tableau Cloud Enhancement* remains in the **Product Backlog** for next-phase continuous deployment.

3. Enterprise Key Performance Indicators (KPIs)

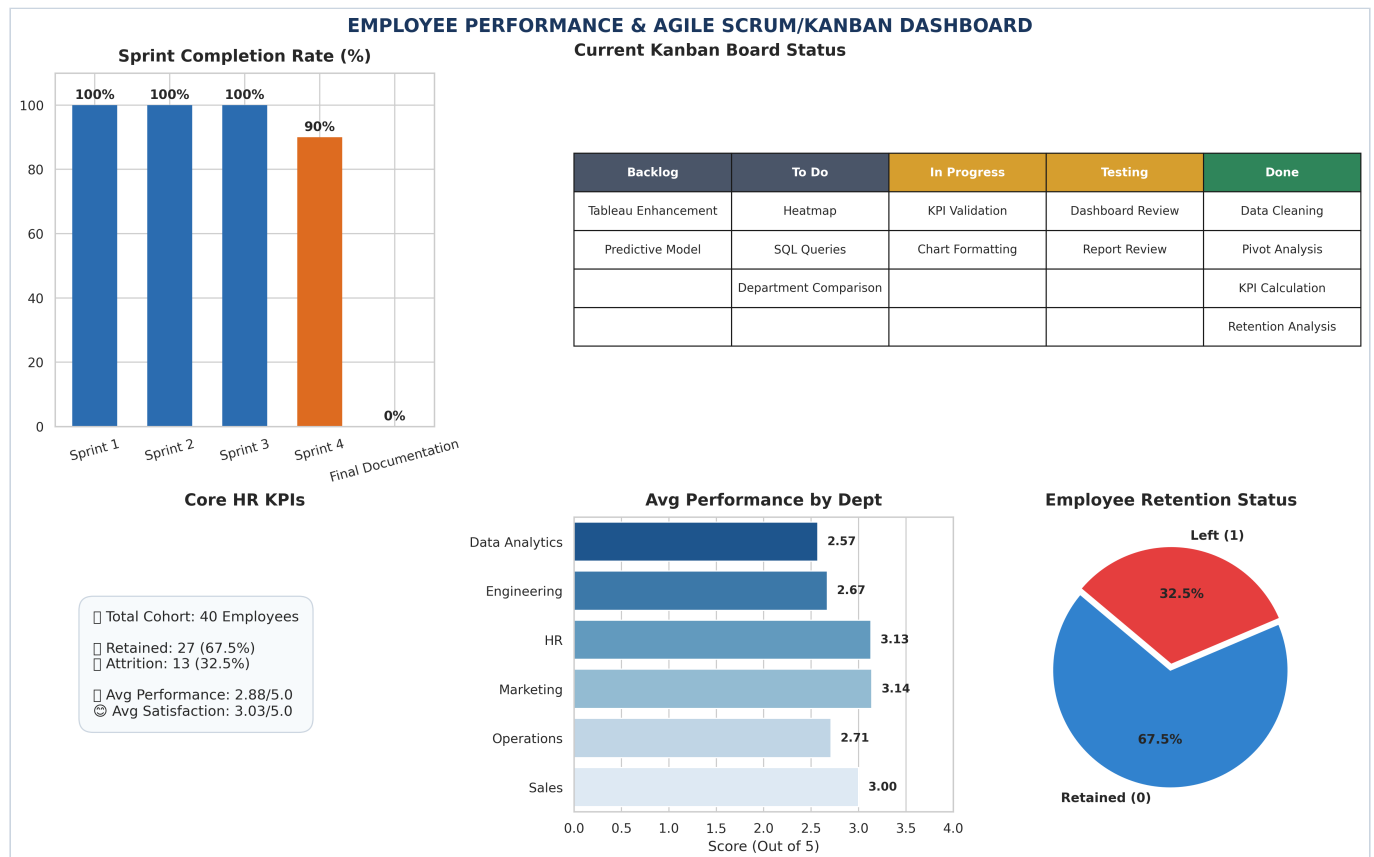
The raw dataset parameters yield a critical snapshot of workforce distribution and productivity scales.

40	67.5%	32.5%	184.3
TOTAL COHORT	RETENTION RATE	ATTRITION RATE	AVG MONTHLY HRS

Department	Staff Count	Avg Performance (5.0)	Avg Satisfaction (5.0)	Avg Monthly Hours	Avg Projects Completed	Attrition Rate (%)
Data Analytics	7	2.57	3.43	191.9	5.29	28.6%
Engineering	6	2.67	3.00	196.7	4.33	50.0%
HR	8	3.13	3.13	191.4	3.75	25.0%
Marketing	7	3.14	2.86	174.0	4.14	42.9%
Operations	7	2.71	1.86	178.3	3.86	42.9%
Sales	5	3.00	4.20	170.6	5.20	0.0%

4. Codec Technologies - Integrated Project Dashboard

The following unified dashboard screenshot links Codec Technologies' Agile management indicators directly with statistical workforce data outcomes.



5. Root-Cause Attrition & Behavioral Insights

Segmenting employees who exited the corporation versus those successfully retained highlights critical behavioral predictors that demand strategic HR intervention:

- **The High-Performer Overwork Pattern:** Employees who voluntarily separated from the enterprise possessed a substantially higher performance rating average (**3.31 / 5.00**) than individuals currently retained (**2.67 / 5.00**). This shows that topmost talent is separating at a higher rate.
- **The Sentiment Baseline Gap:** Departing personnel showed severe drops in internal satisfaction scores, dropping to an average index of **1.62 / 5.00**, while retained staff remain high at **3.70 / 5.00**. This correlates directly with project fatigue and long monthly hours (averaging **187.2 hours** for exiting staff).
- **Structural Risk Pockets:** The **Engineering** segment demonstrates a critical **50.0% attrition rate**, followed closely by **Marketing** and **Operations** at **42.9%**. Conversely, the **Sales** division has completely mitigated churn, retaining 100% of its headcount.

6. Strategic Recommendations & Actions for Codec Technologies

1. **Deploy Targeted Stay-Interviews:** Prioritize structural reviews inside the Engineering and Marketing departments to pinpoint the underlying causes of high employee churn.
2. **Optimize High-Performer Compensation and Well-being:** Establish immediate talent management policies for staff scoring over 3.5/5 on performance scales, addressing their workload levels to mitigate burnout.
3. **Data Analytics Skill Upskilling:** While the Data Analytics department exhibits high project velocity (5.29 projects), its average performance score is low (2.57). Providing target coaching or refining team sprints can boost quality metrics.